Review – Integrals to Infinity

What does it mean for $\int_a^\infty f(x)\,dx$ to converge?

Does $\int_0^\infty xe^{-x^2}\,dx$ converge?

What does it mean for $\int_{-\infty}^b f(x)\,dx$ to converge?

Does $\int_{-\infty}^0 xe^{-x^2}\,dx$ converge?

What does it mean for $\int_{-\infty}^\infty f(x)\,dx$ to converge?

Does $\int_{-\infty}^\infty xe^{-x^2}\,dx$ converge? If so, to what number?

Motivating Example

Evaluate $\int_{-1}^{1} \frac{1}{x^2}\, dx$ using our traditional techniques.

Note that $\frac{1}{x^2} > 0$ for all x. Draw the graph of $f(x) = \frac{1}{x^2}$. What can we conclude about $\int_{-1}^{1} \frac{1}{x^2}\, dx$?

What went wrong? (Also see Worksheet 6-1 Q6.)

Improper Integrals: Discontinuities

Consider $\int_0^1 \frac{1}{\sqrt{x}}\,dx$. Why doesn't the Fundamental Theorem immediately apply to this integral?

Suppose b is a small *positive* number near 0. Compute $\int_b^1 \frac{1}{\sqrt{x}}\,dx$. Draw the graph and shade in the area being computed. Why do we have to assume b is positive?

In order to compute the integral from 0, we need to move b toward 0. We can do this using a limit. Since we need b to be positive, we need to specify that when writing the limit:

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{b\to 0^+} \int_b^1 x^{-\frac{1}{2}}\,dx$$

Complete the work below. Draw the graph and shade in the area.

$$\int_0^1 \frac{1}{\sqrt{x}}\,dx = \lim_{b\to 0^+} \int_b^1 x^{-\frac{1}{2}}\,dx$$

You should have found that despite the fact that the function approaches ∞ as $x \to 0^+$, the area is finite! Just as integrals that are unbounded on the x axis can be finite, so can integrals that are unbounded on the y axis.

Again, consider $\int_{-1}^{1} \frac{1}{x^2}\, dx$. Look back at the graph of this couple of pages back and write why this is actually correctly computed as *the sum of two integrals*, not just one. Draw the graph and label the two areas.

So

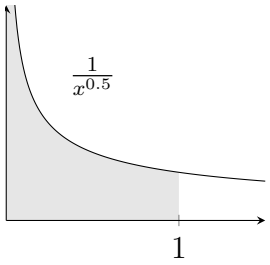
$$\int_{-1}^{1} \frac{1}{x^2}\, dx = \underline{\qquad\qquad} + \underline{\qquad\qquad}$$

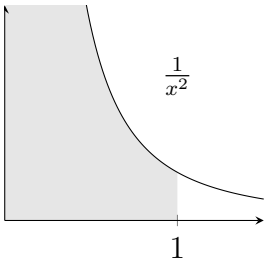
Compute one of those two integrals (you'll need to write it with a limit, of course!).

What do you conclude about the entire integral $\int_{-1}^{1} \frac{1}{x^2}\, dx$, and therefore about the area under the graph of $\frac{1}{x^2}$ between $x = -1$ and $x = 1$?

Reciprocal Power Functions

Here are the graphs of $\frac{1}{\sqrt{x}} = \frac{1}{x^{0.5}}$ and $\frac{1}{x^2}$ plotted on identical axes. Looking at the areas under the graphs between $x = 0$ and $x = 1$, we can see that $\frac{1}{x^{0.5}}$ has a thinner area as $x \to 0^+$ and $y \to \infty$ than $\frac{1}{x^2}$. As we saw above, in fact, the former is finite, and the latter infinite.





This begs the question: for what values of p is $\displaystyle\int_0^1 \frac{1}{x^p}$ finite?

1. Determine whether the following integrals converge or diverge:

□Regrade

(a) $\displaystyle\int_0^1 \frac{1}{x}\,dx$

(b) $\displaystyle\int_0^1 \frac{1}{x^{1.01}}\,dx$

(c) $\displaystyle\int_0^1 \frac{1}{x^{0.99}}\,dx$

(d) Complete the following: $\displaystyle\int_0^1 \frac{1}{x^p}\,dx$ converges if _____. That is, if _____, the area under the graph of $\frac{1}{x^p}$ between $x = 0$ and $x = 1$ is thin enough that despite it being an infinitely 'tall' area, it is finite.

Next, we will look at the other side of these graphs: the area under them between $x = 1$ and $x = \infty$.

2. Determine whether the following integrals converge or diverge:

☐Regrade

(a) $\displaystyle\int_1^\infty \frac{1}{x}\,dx$

(b) $\displaystyle\int_1^\infty \frac{1}{x^{1.01}}\,dx$

(c) $\displaystyle\int_1^\infty \frac{1}{x^{0.99}}\,dx$

(d) Complete the following: $\displaystyle\int_1^\infty \frac{1}{x^p}\,dx$ converges if _____. That is, if _____, the area under the graph of $\frac{1}{x^p}$ between $x = 1$ and $x = \infty$ is thin enough that despite this being an infinitely 'long' area , it is finite.

3. Using your answers above, and noting that $\displaystyle\int_0^\infty f(x)\,dx = \int_0^1 f(x)\,dx + \int_1^\infty f(x)\,dx$, determine for what value(s) of p (if any) $\displaystyle\int_0^\infty \frac{1}{x^p}\,dx$ converges.

☐Regrade

Exercises

Calculate the following integrals:

4. $\int_1^e \frac{1}{x \ln x}\, dx$ (Hint: there's a substitution in there!)
□Regrade

5. $\int_0^4 \frac{8}{x^2 - 16}\, dx$
□Regrade (Hint: start by showing that $\frac{1}{x-4} - \frac{1}{x+4} = \frac{8}{x^2-16}$.)

6. Consider $\int_0^{\frac{\pi}{2}} \tan x \sqrt{\cos x}\, dx$.

☐Regrade

(a) Why is this an improper integral?

(b) Compute this integral. (Hint: Write the integrand in terms of $\sin x$ and $\cos x$. You'll need a substitution.)